

summary

§11.4.108 Four-Layer Verification — Session Commits 457cca4..fab6707

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Scope

Verifies the ~28 commits landed in this session (457cca4 -> fab6707, 81 files changed) against the running boba stack, per §11.4.108 (SOURCE -> ARTIFACT -> RUNTIME-ON-CLEAN-TARGET -> USER-VISIBLE).

Phase 1 — Runtime-impact categorization

git diff --name-only 457cca4..fab6707 enumerated. Every changed file classified:

Category	Files	Runtime impact
C — rebuild required	qBitTorrent-go/internal/jackettapi/health.go, health_test.go	YES — boba-jackett is build: context: ./ qBitTorrent-go, dockerfile: Dockerfile.jackett in docker- compose.yml (not a bind-mount)
B — host-level, non-container	scripts/install-resource-pressure-timer.sh +	NO — systemd -- user timer, entirely

Category	Files	Runtime impact
A — no runtime impact	scripts/systemd/user/boba-resource-pressure-check.{service,timer}	outside the docker-compose stack
	Everything else: docs/**, challenges/**, scripts/commit-push-all.sh, scripts/install-dev-tools.sh, scripts/capture-workable-items-db-delta.sh, scripts/pre_build_verification.sh, scripts/host-power-management/check-no-suspend-calls.sh, constitution (submodule pointer)	NO — dev tooling / challenge scripts / docs / governance, none container-mounted or container-built

Verdict: exactly one runtime-affecting source pair (BOB-112's TTL-cache fix in health.go). No docker-compose.yml, no Dockerfile.jackett, no download-proxy/src/, no plugins/, no config/ changes this session.

Phase 2 — Rebuild + reflash

Finding before acting: the *already-running* boba-jackett container's image (962f4452c725...) was built at 2026-08-18T19:06:17Z — **after** health.go's last touching commit 91b52db (19:02:50Z) — because a prior subagent in this same session (commit e2a2e3e, evidence at docs/qa/BOB-112/) had already rebuilt + RED/GREEN-mutation-verified this exact fix with live wrk load tests (97.1%→0.0% timeout rate, full methodology + numbers in docs/qa/BOB-112/summary.md). Confirmed no commit after 91b52db touched health.go (git log 91b52db..fab6707 -- health.go -> empty).

This task's job was therefore to **redeploy on a genuinely clean target** (§11.4.108 clause 4 — eliminate the stale-deployment/shadow layer by construction) and confirm the fix holds, rather than repeat the prior subagent's RED/GREEN mutation study.

Ran `./start.sh --recreate` (the CLAUDE.md-sanctioned Level-3 restart for “docker-compose.yml, start-proxy.sh, env var, base image” changes — the correct level given a rebuilt image needed a fresh container instance). Full transcript: `start_sh_recreate.log`.

- Tracker-cookie loader ran first (per operator mandate 2026-08-15): all 5 trackers (rutracker/nnmclub/rutor/kinozal/iptorrents) reported `UNCHANGED (idempotent)` — no drift, no re-extraction needed.
- boba-ctl (Go orchestrator) executed `podman-compose ... down then ... up -d` — exit 0 both steps.
- Only boba’s own 4 services (qbittorrent, jackett, qbittorrent-proxy, boba-jackett) were touched. Confirmed via before/after `podman ps -a` (`container_status.log`) that the 15 unrelated containers belonging to other projects on this host (proxy-*, lava-*, helix-*, `deploy_caddy_1`, `wonderful_dijkstra`) were **not restarted, not recreated, not touched** — their Up 2 hours / Up 12 hours uptimes are unchanged across the before/after snapshots.
- All 4 boba containers reached healthy within ~1 minute (qbittorrent/jackett: healthy at 52s; boba-jackett/qbittorrent-proxy: healthy at 46s).
- Post-recreate cookie lengths (byte counts only, §11.4.10 — no values logged) are **byte-identical** to the pre-recreate baseline for all 5 trackers (`cookie_state_post_recreate.log`).

Phase 3 — Runtime verification

All 5 endpoints reachable post-recreate (`endpoint_health.log`, verbatim):

Endpoint	Result
GET :7186/ api/v2/app/ version (qBittorrent via proxy)	200, body v5.2.3
POST :7186/ api/v2/ auth/login (admin/ admin)	204 (qBittorrent’s own success response)
GET :7187/ health	200, {"status":"healthy","service":"merge-search","version":"1.0.0"}

Endpoint	Result
(merge service)	
GET :7189/healthz (boba-jackett, NEW TTL cache)	200, {"status":"ok","db_ok":true,"jackett_ok":true,"version":"0.1.0","upt
GET :9117/api/v2.0/server/config (Jackett upstream)	302 (redirect to admin auth — reachable, expected for an unauthenticated user)

BOB-112 fix — live verification on the fresh container

Artifact layer (artifact_layer_check.txt): the BOB-112 fix’s unique log.Printf format string (added in 91b52db, absent in every prior version of health.go) is present in the **freshly-recreated** container’s binary:

```
boba-jackett: /healthz Jackett cache refresh #%d (hits=%d
misses=%d ratio=%.1f%% ok=%v)
```

Image digest 962f4452c725... — identical to the digest the prior subagent’s RED/GREEN mutation study (docs/qa/BOB-112/summary.md) recorded as the confirmed post-fix, post-revert-verification build. No drift.

Runtime layer (wrk_healthz_evidence.log): brief wrk -t2 -c50 -d5s --timeout 3s --latency http://localhost:7189/healthz against the container that had been up for well under a minute (genuinely cold cache at test start):

```
270085 requests in 5.00s, 48.17MB read
Requests/sec: 54008.16
Latency: avg 1.08ms, p50 844us, p90 2.31ms, p99 4.42ms
```

0 timeouts / 0 socket errors (wrk prints no “Socket errors” line when all three counters — connect/read/write/timeout — are zero) — well under the task’s <5% threshold, and consistent with the prior subagent’s full 30sx100c study (0.0% timeouts, 27k req/s).

Container logs (`podman logs boba-jackett`) show exactly 2 upstream `Jackett.GetCatalog()` calls across the whole test window — one at boot (cache refresh #1, cold cache) and one ~31s later once the 30s TTL first expired — confirming the cache, not luck, is what collapsed 270k / healthz hits into 2 real upstream calls.

Honest gaps / not covered by this task

- **No image rebuild was forced.** `./start.sh --recreate` is `compose down && compose up -d` — it does not pass `--build`, so it reused the existing `boba_boba-jackett:latest` image rather than compiling a fresh one from current HEAD (fab6707). This is not a gap for BOB-112 specifically (proven byte-identical via the format-string + digest match above, and `git log 91b52db..fab6707 -- health.go` is empty so there is nothing newer to rebuild), but it means this task did not independently *reproduce* a build from source — it verified the artifact already known to be correct survives a clean container-level redeploy. A from-scratch `podman-compose build boba-jackett` was not run because it would have produced a byte-identical image to the one already running (no source delta since the prior subagent's own build), and CLAUDE.md's Critical Constraints table has no sanctioned `start.sh` subcommand for “force-rebuild a container's Go source without a compose/Dockerfile/env change” (only `--reload-python`, `--reload-plugins`, and `--recreate` exist) — using a raw `podman-compose build` would have stepped outside the orchestrator-only mandate for a rebuild this task's own evidence shows was unnecessary.
- **§11.4.185 manual QA (human confirmation) not performed** — out of scope for this automated verification subagent; this report is the automated §11.4.108 evidence layer only, not the terminal human sufficiency gate.
- **qbittorrent-proxy (Python/FastAPI merge service) source unchanged this session** — verified reachable (`:7187/health 200`) but not independently re-tested beyond the health endpoint, since no commit touched `download-proxy/src/`.
- The host-level `boba-resource-pressure-check.timer` (BOB-076/task-77) was spot-checked (`systemctl --user list-timers`, active, fired ~5 min prior) for completeness but is **not part of the boba container stack** and its own evidence trail already lives at `docs/qa/task-77/`.

Verdict

PASS — the one runtime-affecting change this session (BOB-112 TTL cache) is confirmed live on a genuinely fresh container instantiation via both artifact-layer (binary string + image digest match) and runtime-layer (0%-timeout wrk run + exact cache-refresh-count log evidence) proof. All 5 boba-stack endpoints reachable post-recreate. Zero unrelated host containers disturbed. All 5 tracker cookies preserved byte-length-identical across the recreate.